

REST API Manual (Hardware & Mobile App)

Digitex School Management System

Version: Laravel 11 | **Document date:** June 10, 2026

E-Digitex SMS — Multi-Institution Education Platform
Confidential — For authorized users only

Digitex SMS — REST API Manual (Hardware & Mobile App)

1. Overview

1.1 Base URL

All schools use one platform address:

```
https://e-digitex.com
```

API base:

```
https://e-digitex.com/api
```

There are **no per-school subdomains**. Login, mobile apps, hardware devices, and webhooks all use **e-digitex.com**.

Example login page: **https://e-digitex.com/login**

All versioned endpoints use prefix `/v1`.

1.2 Response Format

JSON responses unless noted. Standard shapes:

Success:

```
{ "success": true, "data": { ... } }
```

Error:

```
{ "success": false, "message": "Human-readable error" }
```

HTTP status codes: `200` OK, `400` Bad Request, `401` Unauthorized, `403` Forbidden, `404` Not Found, `422` Validation Error, `503` Service Unavailable.

1.3 Common Headers

Header	Value	When
Accept	application/json	All requests
Content-Type	application/json	POST/PUT bodies
Authorization	Bearer {token}	Sanctum-authenticated routes
X-Hardware-Secret	{HARDWARE_SECRET}	Hardware scanner endpoints
X-Institution-Id	{institution_id}	Hardware GET attendance today (required)

2. Authentication

2.1 Mobile Login

POST /v1/login

Auth: None (public)

Body:

```
{
  "email": "user@school.com",
  "password": "secret"
}
```

The `email` field accepts **email**, **username**, or student **shortcode**.

Success (200):

```
{
  "success": true,
  "token": "1|plainTextSanctumToken...",
  "user": {
    "id": 1,
    "name": "John Doe",
    "email": "john@school.com",
    "role": "Teacher",
    "institution_id": 3,
    "school_name": "Example High School",
    "school_logo": "https://..."
  }
}
```

Errors: 401 invalid credentials, 403 inactive account.

Store `token` securely on device. Send on all subsequent requests:

```
Authorization: Bearer 1|plainTextSanctumToken...
```

2.2 Update FCM Token (Push Notifications)

POST `/v1/update-fcm-token`

Auth: Sanctum

Body:

```
{ "fcm_token": "firebase-device-token" }
```

3. User Profile API

Prefix: `/v1/profile` — **Auth:** Sanctum

GET `/v1/profile/`

Returns authenticated user profile (name, email, phone, address, photo URL, role).

POST `/v1/profile/update`

Body (partial allowed):

```
{
  "phone": "+1234567890",
  "address": "123 Street",
  "password": "newpassword",
  "password_confirmation": "newpassword"
}
```

Profile picture: multipart form field `profile_picture` (where supported by client).

4. Hardware API (NFC / RFID / QR Gate)

Prefix: `/v1/hardware`

4.1 Security

- Set `HARDWARE_SECRET` in server `.env` (long random string).
- Every hardware request must include header:

```
X-Hardware-Secret: {same value as HARDWARE_SECRET}
```

- If secret is **not configured**, API returns **503** (disabled).
- Secret comparison uses timing-safe `hash_equals`.
- For read endpoints, also send:

```
X-Institution-Id: 3
```

4.2 Universal Scan

POST `/v1/hardware/attendance/scan`

Auth: `X-Hardware-Secret`

Body:

```
{
  "uid": "A1B2C3D4",
  "device_id": "GATE-01",
  "timestamp": "2026-06-09T08:15:00",
  "method": "rfid",
  "purpose": "attendance"
}
```

Field	Required	Description
uid	Yes	NFC/RFID/QR token or admission number
device_id	No	Scanner identifier for audit
timestamp	No	ISO datetime of scan (default: now)
method	No	rfid , nfc , qr (default: rfid)
purpose	No	See purposes below

Purposes (purpose):

Value	Behavior
attendance (default)	Student/staff check-in or check-out
fee_check	Returns fee balance / payment threshold
pickup	Process pickup QR or NFC gate pickup
report_card	Returns published exam marks summary

UID matching: System searches student fields nfc_tag_uid , rfid_uid , qr_code_token , admission_number and staff equivalents. UIDs are normalized (case/colon variants).

Attendance response (example):

```
{
  "success": true,
  "status": "success",
  "message": "Check-In Recorded",
  "action": "check_in",
  "student_name": "Jane Doe",
  "time": "08:15 AM"
}
```

Fee check: Returns balance breakdown; `success: false` when payment threshold not met (with data payload).

Pickup: Accepts pickup tokens (`PKUP-*` , `QR-*`) or student NFC tap; sets status to `scanned` and notifies class teacher.

Report card: Only **published** exam results returned; may block if institution has debt blocking enabled.

4.3 Today's Attendance List

GET `/v1/hardware/attendance/today`

Auth: `X-Hardware-Secret` + `X-Institution-Id` **OR** Sanctum user token

Hardware headers:

```
X-Hardware-Secret: ...
X-Institution-Id: 3
```

Sanctum: Results scoped by role (student → self, parent → children, teacher → classes, admin → institution).

Response:

```
{
  "success": true,
  "data": [
    {
      "id": 101,
      "student_name": "Jane Doe (ADM-001)",
      "admission_no": "ADM-001",
      "photo": "https://.../storage/students/...",
      "time": "08:15 AM",
      "time_in": "08:15 AM",
      "time_out": "--:--",
      "action": "Arrival",
      "status": "Present",
      "status_color": "#10B981",
      "action_color": "#3B82F6"
    }
  ]
}
```

4.4 Teacher Class Absentees

GET /v1/hardware/attendance/absentees

Auth: Sanctum (Teacher)

Returns absentee report for teacher's assigned classes (multi-day structure). Used by teacher mobile dashboard.

5. Staff / Teacher Pickup API

Prefix: /v1/pickup — **Auth:** Sanctum

Blocked roles: Student, Guardian (receive empty or 403).

5.1 Pending Count

GET /v1/pickup/count

```
{ "success": true, "count": 3 }
```

Scoped by institution and teacher's classes.

5.2 Pending Pickup List

GET /v1/pickup/pending

```
{
  "success": true,
  "data": [
    {
      "pickup_id": 55,
      "student_name": "Jane Doe (ADM-001)",
      "admission_number": "ADM-001",
      "parent_phone": "+1234567890",
      "class_name": "Grade 5 A",
      "requested_by": "Parent App",
      "status": "scanned",
      "scanned_by_device": "GATE-01",
      "time": "14:30"
    }
  ]
}
```

5.3 Approve Pickup

POST /v1/pickup/approve

Body:

```
{ "pickup_id": 55 }
```

Authorization: Teacher (student in assigned class), School Admin, Head Officer (same institution), Super Admin.

Success:

```
{ "success": true, "message": "pickup_approved_success" }
```

Triggers parent SMS/WhatsApp if preferences enabled.

5.4 Generate OTP (Gate fallback)

POST /v1/pickup/generate-otp

Auth: Sanctum — Guardian, Staff, Admin roles (not Student)

Body:

```
{ "student_id": 42 }
```

`student_id` may be numeric ID or admission number. Guardian must be linked to student; staff/admin must belong to same institution.

Success:

```
{  
  "success": true,  
  "message": "pickup_otp_sent",  
  "student_id": 42  
}
```

OTP sent via SMS to registered parent phone; cached 15 minutes.

5.5 Verify OTP

POST `/v1/pickup/verify-otp`

Body:

```
{  
  "student_id": 42,  
  "otp": "123456"  
}
```

Creates pickup record with status `scanned`. Returns wait-for-teacher message.

6. Student Portal API

Prefix: `/v1/student` — **Auth:** Sanctum

Allowed roles: Student (own record) or Guardian (first linked child).

6.1 Attendance History

GET `/v1/student/attendance`

Last 30 records with date, subject, status, check-in/out times.

6.2 Fees & Invoices

GET /v1/student/fees

```
{
  "success": true,
  "data": {
    "total_fees": 5000.00,
    "paid_fees": 3000.00,
    "outstanding": 2000.00,
    "invoices": [
      {
        "id": 10,
        "invoice_number": "INV-2025-001",
        "total_amount": 2500.00,
        "paid_amount": 1000.00,
        "status": "partial",
        "due_date": "2025-09-01"
      }
    ]
  }
}
```

6.3 Homework

GET /v1/student/homework

Assignments for enrolled class (typically 7-day window) with subject, title, deadline.

6.4 Results

GET /v1/student/results

Published exam records only. Returns 403 /empty if block_reports_on_debt enabled and student has outstanding fees.

6.5 Gate Pass (Pickup QR)

POST /v1/student/gate-pass

Generates pickup token/QR for authenticated student (or guardian's child). Returns token string and expiry for QR encoding.

6.6 Submit Request

POST `/v1/student/requests`

Body:

```
{
  "type": "absence",
  "reason": "Medical appointment",
  "start_date": "2026-06-10",
  "end_date": "2026-06-10"
}
```

Types: `absence`, `late`, `sick`, `early_exit`, `leave`, `other`. Creates pending student request ticket.

7. Chatbot API

Prefix: `/v1/chatbot`

7.1 Webhook (Inbound Messages)

POST `/v1/chatbot/webhook/{provider}`

Auth: None (public endpoint — secure via provider signatures in production)

Providers: `infobip`, `twilio`, `meta`, `mobishastra`

Provider POSTs message payload; system normalizes to `{ from, to, body, provider }` and passes to `ChatbotLogicService`.

Configure in provider dashboard:

```
https://e-digitex.com/api/v1/chatbot/webhook/twilio
```

7.2 Generate Pickup QR (Authenticated)

POST `/v1/chatbot/generate-qr`

Auth: Sanctum

Body:

```
{
  "student_id": 42,
  "otp": "123456",
  "requester_name": "Parent Name"
}
```

Verifies OTP then returns pickup QR/token data for WhatsApp flow integration.

8. Error Codes Reference

HTTP	Meaning
401	Missing/invalid token or hardware secret
403	Role not allowed or institution mismatch
404	Student/pickup/record not found
422	Validation failed
503	Hardware API disabled (no HARDWARE_SECRET)

9. Hardware Integration Guide

9.1 Recommended Setup

- Generate strong `HARDWARE_SECRET` (32+ chars).
- Configure each gate device with secret and institution ID.
- Register student/staff UIDs in admin panel before go-live.
- Set school timings in Configuration → School year (`school_start_time` , `late_margin_time` , `double_tap_wait_time`).
- Enable notification events: `student_arrival` , `student_departure` in Configuration.

9.2 Sample cURL — Attendance Scan

```
curl -X POST "https://e-digitex.com/api/v1/hardware/attendance/scan" \
-H "Content-Type: application/json" \
-H "X-Hardware-Secret: YOUR_SECRET" \
-d '{"uid":"ADM-001","device_id":"GATE-1","purpose":"attendance","method":"rfid"}'
```

9.3 Sample cURL — Today's List

```
curl "https://e-digitex.com/api/v1/hardware/attendance/today" \
-H "X-Hardware-Secret: YOUR_SECRET" \
-H "X-Institution-Id: 3" \
-H "Accept: application/json"
```

9.4 Sample cURL — Mobile Login

```
curl -X POST "https://e-digitex.com/api/v1/login" \
-H "Content-Type: application/json" \
-d '{"email":"teacher@school.com","password":"secret"}'
```

9.5 Sample cURL — Authenticated Request

```
curl "https://e-digitex.com/api/v1/pickup/pending" \
-H "Authorization: Bearer 1|token..." \
-H "Accept: application/json"
```

10. Mobile App Integration Notes

10.1 Token Lifecycle

- Token issued at login; no refresh endpoint documented — re-login on 401.
- Store token in secure storage (Keychain/Keystore).
- Call `/v1/update-fcm-token` after login for push notifications.

10.2 Role-Based UI

Use `user.role` from login response to show:

Role	Primary API surfaces
Teacher	/v1/pickup/* , /v1/hardware/attendance/absentees
Student	/v1/student/*
Guardian	/v1/student/* (child context)
Admin	Broader web app; limited mobile pickup approval

10.3 Pickup State Machine

pending → scanned → approved | rejected

Mobile teacher app lists `pending` and `scanned`; approve moves to `approved` and notifies parent.

10.4 Offline / Retry

Hardware scanners should retry POST on network failure with same `uid` + `timestamp`; server deduplicates check-in/out using double-tap wait setting.

11. Rate Limiting & Production

- Configure Laravel rate limiter for `/v1/login` to prevent brute force.
- Use HTTPS only in production.
- Rotate `HARDWARE_SECRET` if compromised; update all devices.
- Monitor `storage/logs/laravel.log` for hardware `Unknown UID` warnings.

12. Endpoint Summary Table

Method	Endpoint	Auth
POST	/v1/login	Public
POST	/v1/update-fcm-token	Sanctum
GET	/v1/profile/	Sanctum

POST	/v1/profile/update	Sanctum
POST	/v1/hardware/attendance/scan	Hardware secret
GET	/v1/hardware/attendance/today	Hardware secret + Institution ID OR Sanctum
GET	/v1/hardware/attendance/absentees	Sanctum
GET	/v1/pickup/count	Sanctum
GET	/v1/pickup/pending	Sanctum
POST	/v1/pickup/approve	Sanctum
POST	/v1/pickup/generate-otp	Sanctum
POST	/v1/pickup/verify-otp	Sanctum
GET	/v1/student/attendance	Sanctum
GET	/v1/student/fees	Sanctum
GET	/v1/student/homework	Sanctum
GET	/v1/student/results	Sanctum
POST	/v1/student/gate-pass	Sanctum
POST	/v1/student/requests	Sanctum
POST	/v1/chatbot/webhook/{provider}	Public
POST	/v1/chatbot/generate-qr	Sanctum

End of REST API Manual